Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

Group: 25-26J-138

Problem Statement and Stakeholders

Context

- University environments are increasingly experiencing high academic pressure, emotional stress, and cognitive overload among undergraduate students, while access to timely and personalized mental health support remains limited.

Stakeholders

Primary Users

- Undergraduate students experiencing academic stress

Secondary Users

- University counsellors and mental health professionals

Decision Makers

- University administrators and academic institutions responsible for student well-being programs

Problem Statement and Stakeholders

Observed Issue

- Increasing levels of stress and cognitive overload among students
- Limited availability of scalable and real-time emotional support systems
- Existing solutions lack personalization and explainability

Why It Matters

- Stress negatively affects academic performance and student retention
- Delayed support increases mental health risks
- Universities require scalable AI-driven solutions to support large student populations

Research Gap and Objectives (Recap)

Research Gap

Existing AI chatbots:

- Provide generic responses
- Lack explainable reasoning
- Do not adapt to user emotional patterns over time

Main Objective

- Develop an AI-powered chatbot system that detects student stress, analyzes cognitive load, and provides personalized interventions using neuro-symbolic reasoning and continual learning.

Solution Overview

Proposed System

- An AI chatbot platform that detects stress and cognitive load in undergraduate students and provides adaptive emotional support.

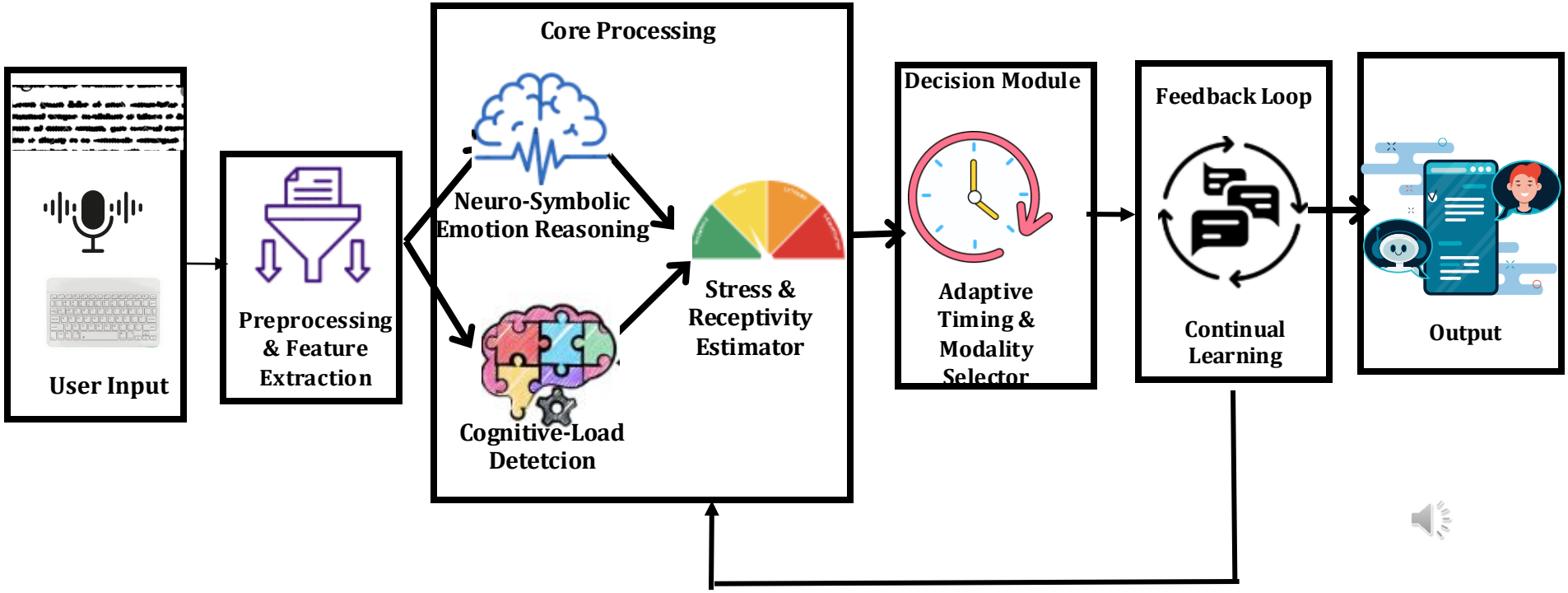
Key Features

- Real-time stress detection from voice and text
- Cognitive load detection using behavioural signals
- Neuro-symbolic reasoning engine
- Emotion-aware continual learning
- Adaptive intervention system

Current Completion Status

- Approximately 85–90% completed

Overall System Architecture and Integration



System Architecture & Integration

System Component	Team Member
Neuro-Symbolic emotion Reasoning Engine	**IT22003928** F N F Ramla
Cognitive load detection via Passive behavioral signals	**IT22332912** Rathnayake R M C T
Emotion – Aware Continual Learning Engine with Personalized Trends Visualization	**IT22886118** Dahanaka D.M.C.B
Adaptive Intervention Timing & Modality Selector	**IT22628404** Safwa M.R.F

System Architecture & Integration

Integration Points

- Python ML models
- Sensor data via Arduino serial communication
- REST APIs for data exchange
- JSON data format

Deployment Setup

- Edge AI processing
- Local server
- Sensor integration

Commercialization / Impact

Target Users

- Universities
- Undergraduate students
- Counseling departments

Value Proposition

- Explainable AI stress detection
- Personalized emotional support
- Adaptive intervention system

Deployment Potential

- University mental health platforms
- Student wellbeing apps
- Academic support systems

DEMO PLAN

- Step 1 :Real-Time Multimodal Stress Detection Module – Live Demo + Performance Improvement

Model	Accuracy	Precision (Neg/Pos)	Recall (Neg/Pos)	F1 (Neg/Pos)
Decision Tree	80.8%	0.78 / 0.84	0.85 / 0.76	0.82 / 0.80
Logistic Regression	68.9%	0.66 / 0.73	0.78 / 0.59	0.72 / 0.66
SVM	73.1%	0.68 / 0.82	0.87 / 0.59	0.76 / 0.69
XGBoost	94.6%	0.91 / 0.97	0.97 / 0.92	0.94 / 0.94

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voice model - svc

	precision	recall	f1-score	support
0	0.64	0.32	0.42	22
1	0.77	0.93	0.84	54
accuracy			0.75	76
macro avg	0.70	0.62	0.63	76
weighted avg	0.73	0.75	0.72	76

xgboost

	precision	recall	f1-score	support
0	0.4176	0.3455	0.3781	110
1	0.7491	0.8022	0.7748	268
accuracy			0.6693	378
macro avg	0.5834	0.5738	0.5764	378
weighted avg	0.6526	0.6693	0.6593	378

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Classification Report:

	precision	recall	f1-score	support
0	0.5000	0.1154	0.1875	26
1	0.6761	0.9412	0.7869	51
accuracy			0.6623	77
macro avg	0.5880	0.5283	0.4872	77
weighted avg	0.6166	0.6623	0.5845	77

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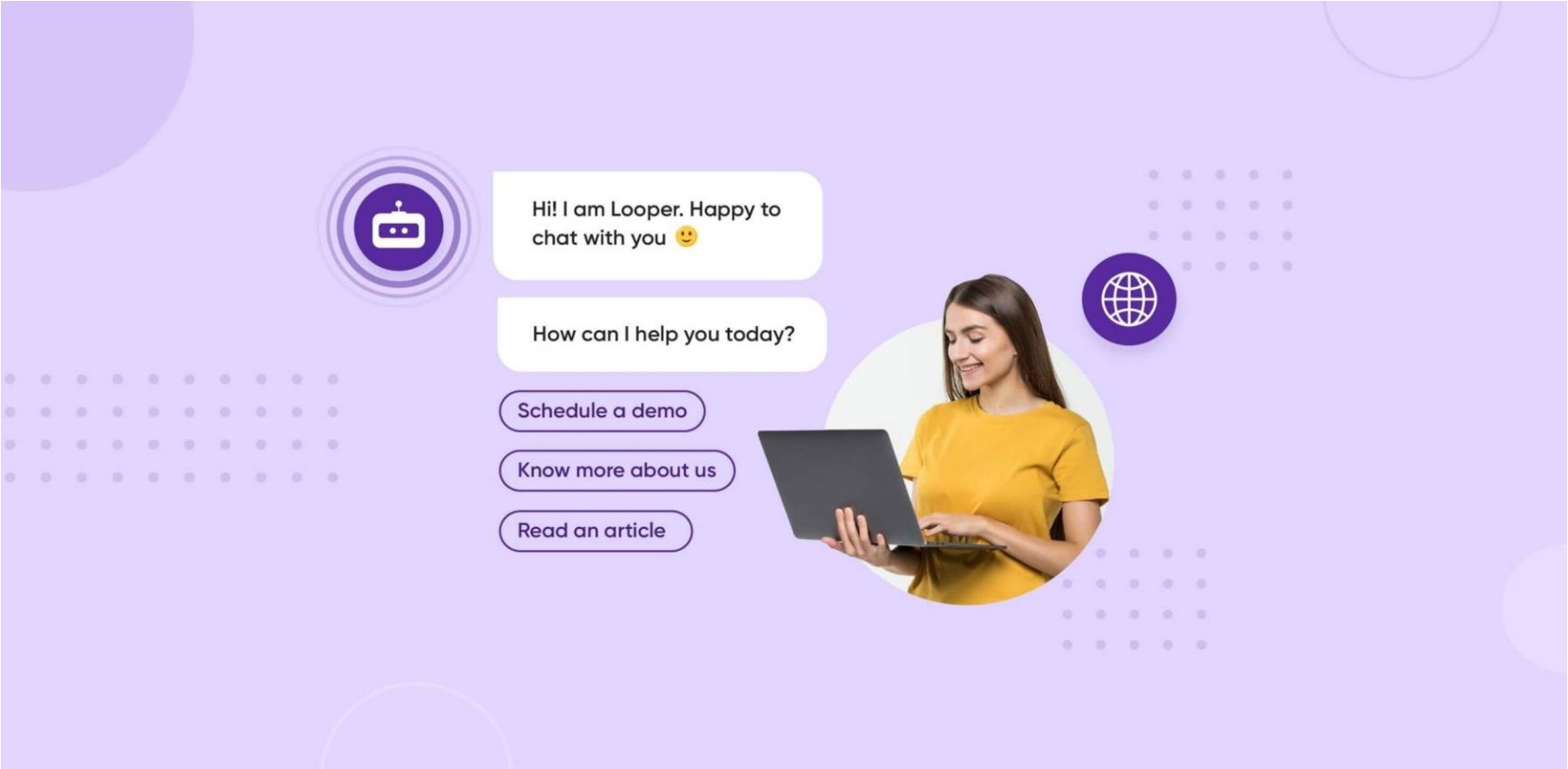
- Step 2 :Passive Behavioral Signal Monitoring Module – Live Demo + Performance Improvement
- Step 3 :Trends Visualization live demo + Performance Improvement
- Step 4 :AITMS live demo + Performance Improvement